

Getting Started with the MinnowBoard

With the embedded market being flooded with open source hardware boards, it's tough finding one that not only suits your budget, computing power needs, etc, but is also backed by a strong community, beginner friendly documentation and a great out-of-the-box experience. The author's search finally ended at MinnowBoard, which successfully lived up to all his expectations, from a hobbyist's point of view.



MinnowBoard (see Figure 1) is an open source hardware board based on the Intel x86 architecture. It is also a reference platform for the Yocto Project (<https://www.yoctoproject.org>), which lets you build your own embedded Linux distribution. MinnowBoard can be complemented with plug-in boards called 'lures', which augment its functionality. Listed below are some of the cool features of MinnowBoard.

Processor

- 1GHz Intel Atom E640 processor
- 1GB DDR2 RAM

Connectivity

- SPI, GPIO, I2C, CAN, SATA, SDIO, PCI Express, microSD card slot, DVI
- Gigabit Ethernet
- USB host (x2)

Software

- 4GB microSD card with embedded Linux distribution (the

Angstrom OS)

Size

- 10.66 cm x 10.66 cm (4.2" x 4.2")

Resources

While playing around with MinnowBoard, you will have a number of doubts. The following links would be the best places to start looking for answers:

Websites:

<http://www.elinux.org/Minnowboard>

<http://www.minnowboard.org/>

IRC:

#minnowboard

Operating systems supported by MinnowBoard

At the time of writing this article, Angstrom is the officially supported operating system. Support for Android, Ubuntu and others will follow soon.

Where to buy MinnowBoard

You can buy it from various worldwide distributors such as Mouser, DigiKey, Special Computing, Element14, etc.

The cost of MinnowBoard is currently US\$ 199.

MinnowBoard lures

As mentioned earlier, a 'lure' is a plug-in board (similar in concept to the Arduino 'shield') used to extend the functionalities of MinnowBoard. At the time of writing this article, breakout board (aka 'BOB' lure), trainer lure and beacon lure were available. More are to follow soon. For more information about the lures, please refer to the following link: http://www.elinux.org/Minnowboard:Lures_Specifications

Box contents

When you purchase a MinnowBoard, you get the following items included in the box: MinnowBoard, a USB cable, 4GB microSD card and a power adapter.

Download the latest Angstrom image

Please visit the link below to download the latest Angstrom Linux image for the MinnowBoard: